

11 (long)

(a) Describe the following concepts used in fracture mechanics:

- (i) stress intensity factor K ; [2]
- (ii) fracture toughness K_{IC} ; [2]
- (iii) fast fracture. [2]

(b) An aircraft component is manufactured from an aluminium alloy with a fracture toughness of $28 \text{ MPa m}^{1/2}$. Testing has established that fracture occurs at a remote stress of 260 MPa when the maximum internal crack length is 2.4 mm. For this same component, alloy and crack orientation, determine whether fast fracture will occur at a stress level of 340 MPa when the maximum internal crack is 1.2 mm. Justify your answer. [7]

(c) A steel girder undergoes sinusoidal cyclic loading between 225 MPa (tension) and 60 MPa (compression). Fatigue crack growth follows the Paris law with exponent $n = 2.5$ and constant $1.5 \times 10^{-10} \text{ MN}^{-n} \text{ m}^{1+1.5n}$. The steel used has a fracture toughness $K_{IC} = 95 \text{ MPa m}^{1/2}$. The largest surface crack present, identified by X-ray inspection, is 2.5 mm in length.

- (i) Calculate the fatigue life of the girder. In the expression for the stress intensity factor, assume that the dimensionless constant Y is equal to 1. [8]
- (ii) Find the decrease in stress range necessary to increase the fatigue life of the girder, estimated in (i), by a factor of 5. [6]
- (iii) It is found that in service the loading amplitude fluctuates with a fraction of the runs being at a lower load amplitude. How would you calculate the effect this has on a fatigue life? [3]